

Answer Key — Biology Grade 11

Sub-Strand 1.1: Taxonomy I • Lesson 8: Classification into the Five Kingdoms and Importance of Taxonomy: Our Final Explanation

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
D	C	D	C	D	A	B	C

USE AFTER: OBSERVE PHASE — Expert groups: evidence table for Group A (Monera and Protocista: Amoeba, Spirogyra, Paramecium)

Q1. A single-celled organism from a pond has a nucleus, moves using temporary extensions of its body (pseudopodia) and takes in food particles. Which kingdom does it belong to?

- A. Fungi
- B. Plantae
- C. Monera
- D. Protocista**

Correct answer: D. A unicellular organism with a nucleus is eukaryotic, so it cannot be in Monera (prokaryotes). Amoeba-type organisms belong to Kingdom Protocista.

USE AFTER: OBSERVE PHASE — Group B (Fungi: Rhizopus, Agaricus mushroom, yeast) card sort into the five kingdom envelopes

Q2. A mushroom grows in soil. It has no chlorophyll and absorbs food from decaying matter. Which kingdom does it belong to, and why?

- A. Animalia, because it cannot make its own food
- B. Plantae, because it grows in soil
- C. Fungi, because it has no chlorophyll and absorbs nutrients from its surroundings**
- D. Protocista, because it has no leaves or roots

Correct answer: C. Fungi are eukaryotic, non-photosynthetic and absorb food from their surroundings. Growing in soil does not make an organism a plant.

USE AFTER: OBSERVE PHASE — Hierarchy card arrangement for maize: writing the binomial name Zea mays correctly

Q3. Which of these is the correct way to write the binomial name of maize?

- A. Zea Mays
- B. zea Mays
- C. zea mays
- D. Zea mays**

Correct answer: D. In binomial nomenclature the genus name starts with a capital letter and the species name is entirely in lowercase. The name is italicised, or underlined when handwritten.

USE AFTER: OBSERVE PHASE — Hierarchy card arrangement (Kingdom, Phylum, Class, Order, Family, Genus, Species) for maize

Q4. Which list shows the taxonomic ranks in the correct order, from the largest group to the smallest group?

- A. Kingdom, Class, Phylum, Order, Family, Genus, Species
- B. Kingdom, Phylum, Order, Class, Family, Genus, Species
- C. Kingdom, Phylum, Class, Order, Family, Genus, Species**
- D. Kingdom, Phylum, Class, Order, Genus, Family, Species

Correct answer: C. The correct hierarchy is Kingdom, Phylum, Class, Order, Family, Genus, Species. Each rank is nested inside the one before it.

USE AFTER: EXPLAIN PHASE — Class discussion and debate: 'Is algae (Spirogyra) a plant?' using Claim-Evidence-Reasoning

Q5. A learner says, "Spirogyra is an animal because it lives in water." Which response is the best scientific reply?

- A. Correct, because all organisms in water are animals.
- B. Correct, because Spirogyra is too small to be a plant.
- C. Wrong, because anything green must be a plant.
- D. Habitat is not a reliable rule. We should use observable features such as cell type, chlorophyll and mode of nutrition.**

Correct answer: D. Classification into kingdoms depends on shared observable features such as cell type, nutrition and chlorophyll, not on where an organism lives. This is why habitat-based sorting from Lesson 1 was not enough.

USE AFTER: EXPLAIN PHASE — Connecting classification, nomenclature and taxonomy: 'Why did we need one scientific name when mahindi, bhutta and corn already exist?'

Q6. Maize is called mahindi in Kiswahili, bhutta in some communities and corn in English. Why do scientists also give it one scientific name, Zea mays?

- A. One two-part name can be understood by scientists everywhere, whatever the local name.**
- B. Local names are wrong and must be replaced.
- C. The scientific name tells where the plant grows.
- D. Scientific names are always shorter than local names.

Correct answer: A. Binomial nomenclature gives each organism one universal two-part name. This lets people who use different local names communicate about the same organism.

USE AFTER: EXPLAIN PHASE — Discussion of the importance of taxonomy: conserving biodiversity such as Mau Forest, Kakamega Forest and Mt Kenya

Q7. Why is correct identification and naming of organisms important for conserving biodiversity in forests such as Kakamega Forest?

- A. It stops species from changing over time.
- B. It lets scientists know which species are present and which are rare, so they can protect them.**
- C. It makes all species belong to the same kingdom.
- D. It means only the species useful to people need to be protected.

Correct answer: B. Taxonomy identifies and names species so that scientists can record them, find out which are rare or threatened, and plan protection. Conservation does not depend only on how useful a species is.

USE AFTER: MODEL BUILDING PHASE — Final Explanation: 'How can living organisms be systematically organised, named and identified?' (Classification + Naming + Identification = Taxonomy)

Q8. Which statement best describes taxonomy?

- A. Giving organisms local names in different languages
- B. Grouping organisms only by where they live
- C. The science of classifying, naming and identifying living organisms**
- D. The study of how organisms interact with their environment

Correct answer: C. Taxonomy is the science that groups organisms (classification), gives them universal two-part names (nomenclature) and helps identify them using observable characteristics.